

RESULTS AND DISCUSSION

The main objective behind the development of an autonomous car is to reduce the cost of separate functions by integrating all of them into one model.

To achieve the implementation of this model, the major components used are: a Raspberry Pi with SD card pre-installed with Raspbian OS, a breadboard to connect together all the components, an L293 or SN755410 motor driver, Jumper cables to connect everything up, one or two DC motors rated for 6v, 4x AA batteries and holder, HC-SR04 Ultrasonic Sensors, 2 Resistors, RPI Camera to capture images.

The following algorithm depicts the exact working of our autonomous model:

- Live feed is taken from the RPI camera.
- This input is fed to the RPI board connected to the breadboard.
- The input video is then converted into a number of frames and each frame is fed into the algorithm.
- Each picture then goes through the Haar cascade algorithm and checks for stop sign.
- If yes, the car stops.
Else, the algorithm checks the distance between the car and the obstacle via the ultrasonic sensor.
- If detected, then the car stops.
Else, the pictures are fed into the CNN algorithm which tells the car to follow the lane rules.

The motors are controlled by the RPI and L293 chip. RPI gives command to the L293 chip. The above algorithm shows the working of our car model. Once the car has started, the RPI camera gives continuous inputs of pictures which are fed to the RPI board to take respective decisions. These video inputs are divided into frames and then fed to the algorithm. The Haar cascade algorithm is used to

determine the start and stop signs in implementation of such models. If the obstacle is right in front of the car, then the algorithm gives stop sign. If not then the Haar cascade algorithm checks the distance between the car and the obstacle with the help of the ultrasonic sensor. If the obstacle is detected the car stops or else it is given a command to continue moving by following the lane rules.

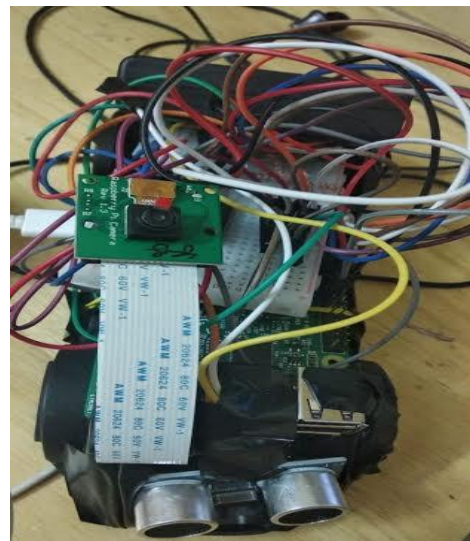


Figure 2: Working model of autonomous car.

The above figure shows the working model of the autonomous car. In order to be able to function efficiently, the car has to be trained with data sets so that it becomes capable of taking its own decisions in the time of crisis.

CONCLUSION

The autonomous car was once implemented into the market to be utilized by users which provided a lot of advantages. It provides safety to the needy so that number of accidents can be reduced. Self-driving vehicles can help the general population who can't drive themselves, for example, the old or debilitated. Open, moderate, and advantageous self-driving vehicles could build the complete number of kilometres driven every year. Other industries such as